

JKBOSE 2026

CLASS 11TH

#XUVV♥

Bhaichara
ON
TOP.

ORGANIC CHEMISTRY:

SOME BASIC PRINCIPLES AND TECHNIQUES

LECTURE-05

CHEMISTRY

VISHAL SIR



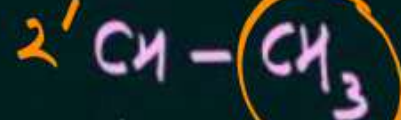
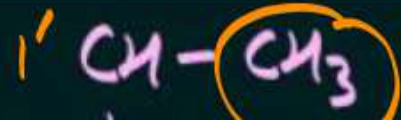
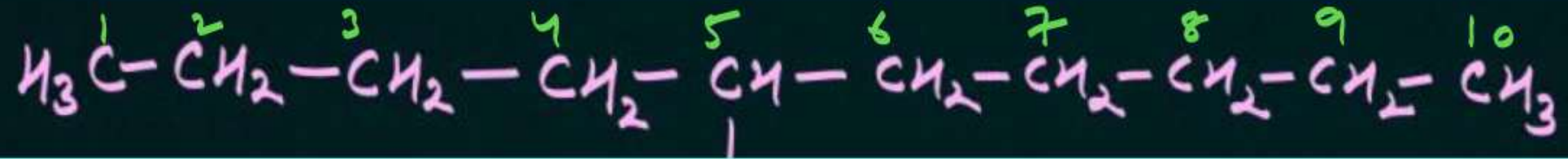
JKBose Wallah

Topics To Covered

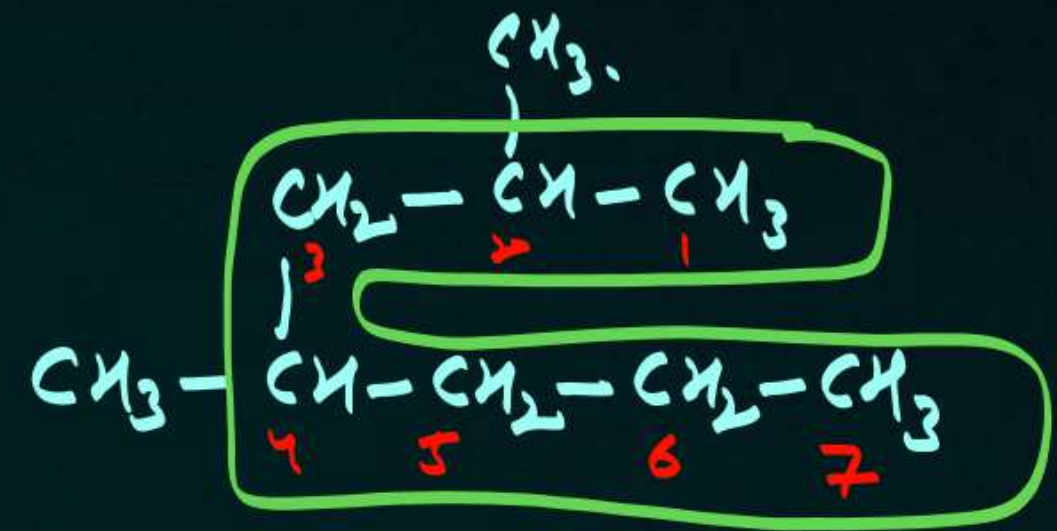
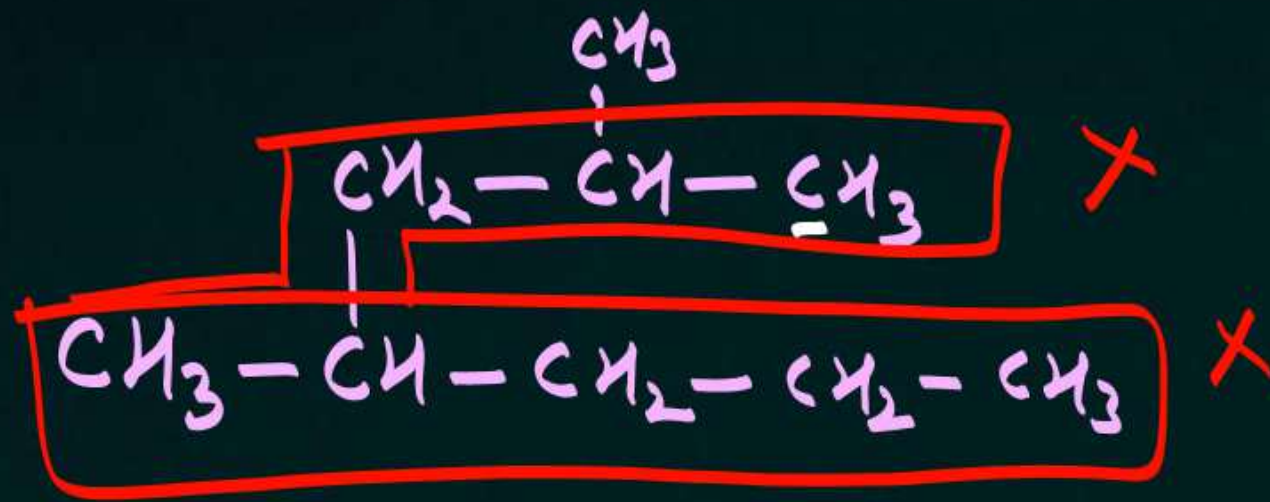
1

IUPAC Nomenclature

Homework

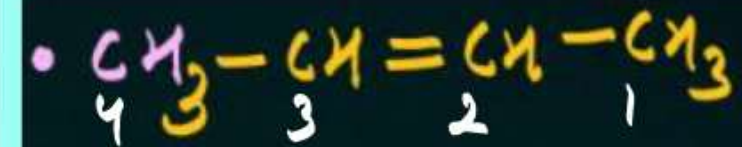
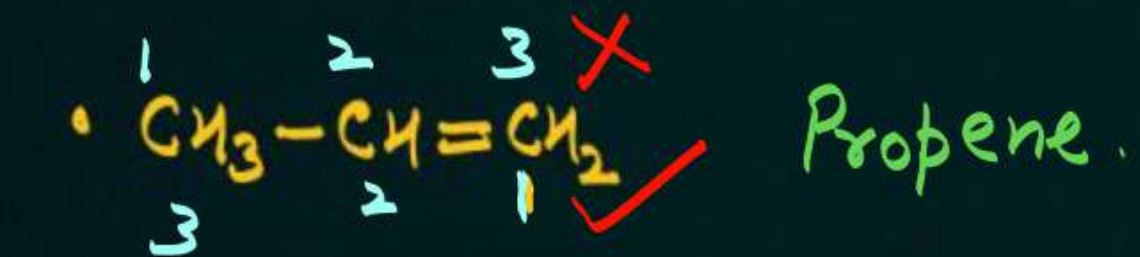
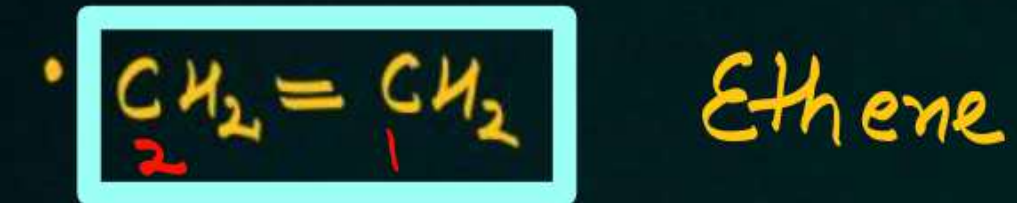


5-(1',2'-Dimethyl propyl) decane.

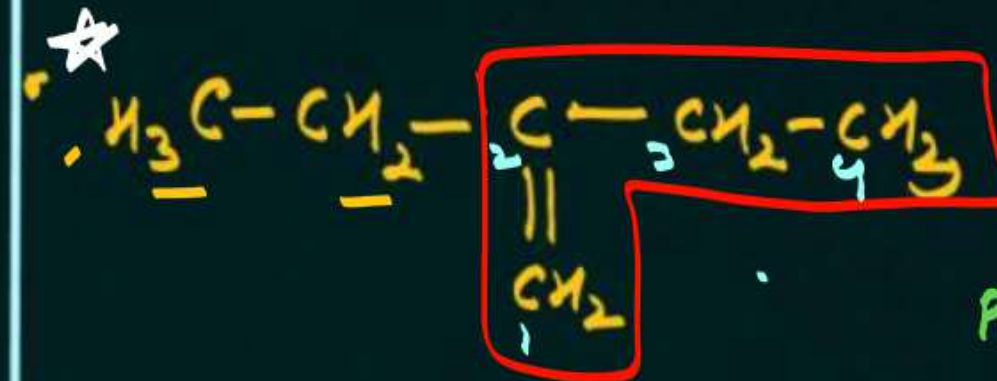


2, 4-Dimethylheptane.

2) NOMENCLATURE OF ALKENES AND ALKYNES.

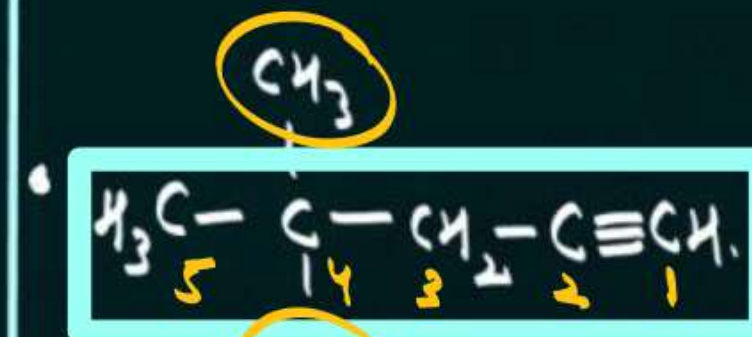


But-2-ene.



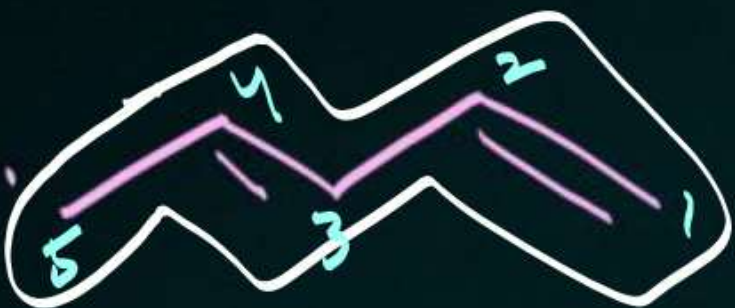
2-Ethyl but-1-ene

PCC = FG > MB > Max no. of C
No = FG > MB >



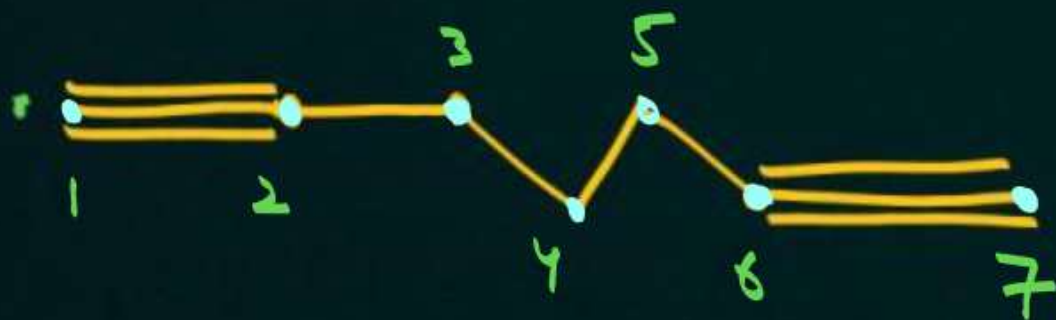
4,4-Dimethyl pentyne.

IUPAC Nomenclature

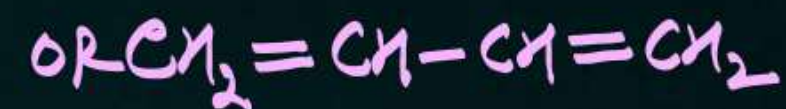


Penta-1,3-diene.

In case of dienes or
dienes we add 'a' with
a word root.



Hepta-1,6-diene



Buta-1,3-diene

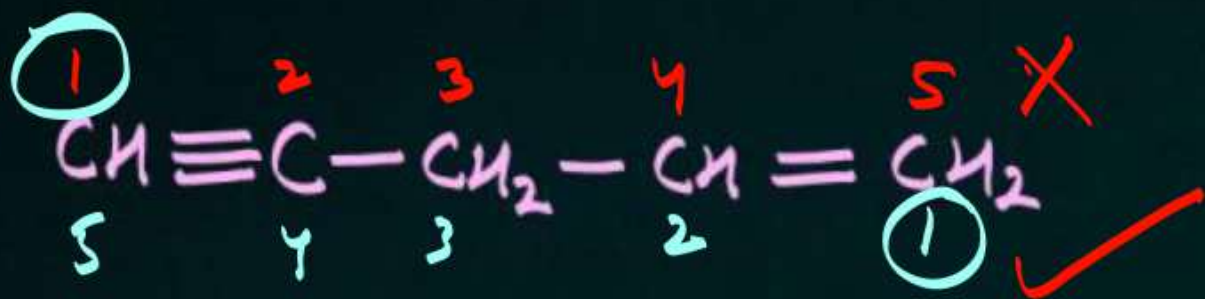


Octa-1,4,7-triene



Octa-1,3,7-triene

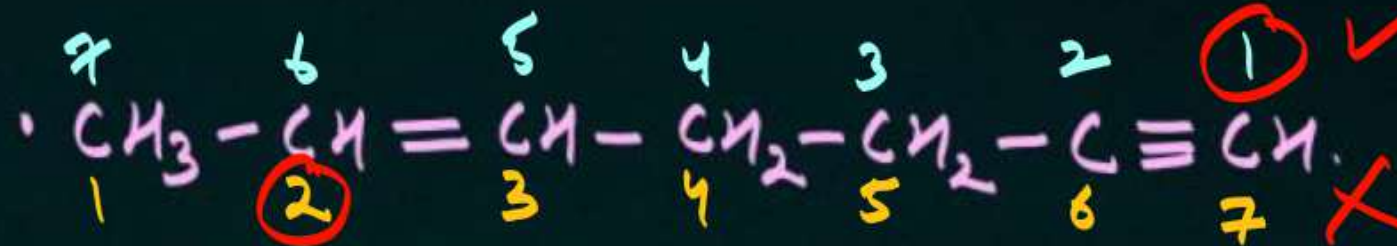
IUPAC Nomenclature



Pent-1-en-4-yne

If there is tie in numbering of ene & yne, then priority (=) > (≡).

ene > yne.



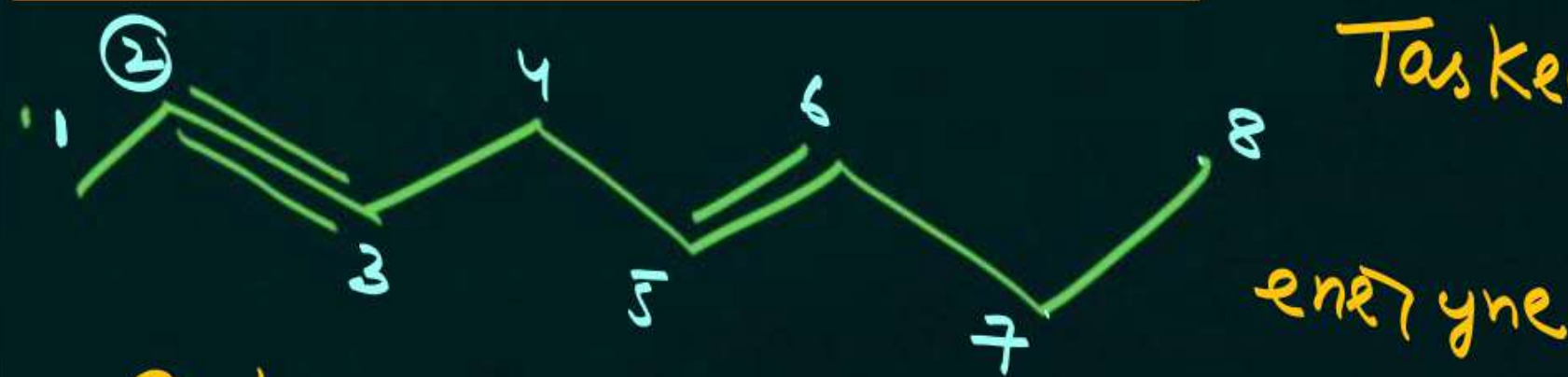
Hept-5-en-1-yne

In case of suffix, if two vowels comes together (a, e, i, o, u, y) then we omit first vowel.

ene > yne

ene > yne

Taskeen.



Oct-5-en-2-yne

ene > yne



Subst + cyclo + wk + ene or alkene or alkyne + FG.



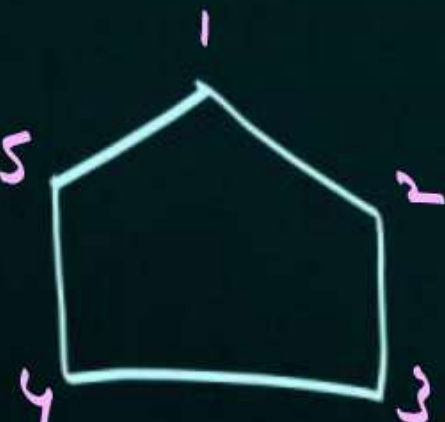
3) Nomenclature of cycloalkanes, cycloalkenes and cycloalkynes.



Cyclopropane



Cyclobutane



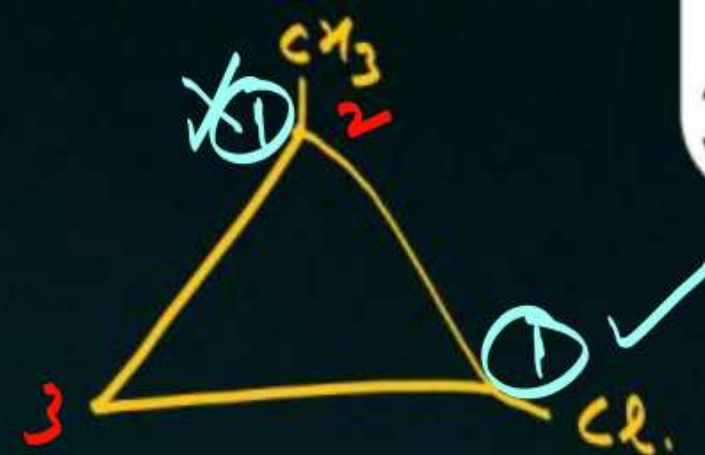
Cyclopentane



Cyclohexane.



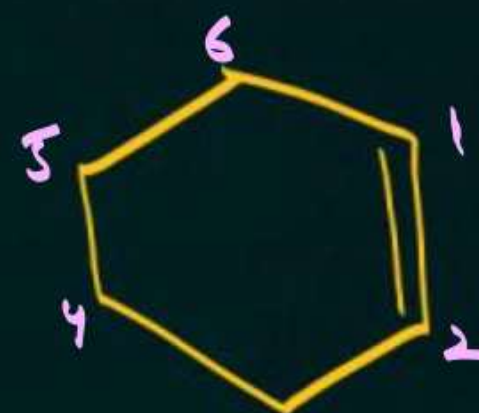
1-Methylcyclopropane



1-Chloro-2-methylcyclopropane.



3-Chlorocycloprop-1-ene.



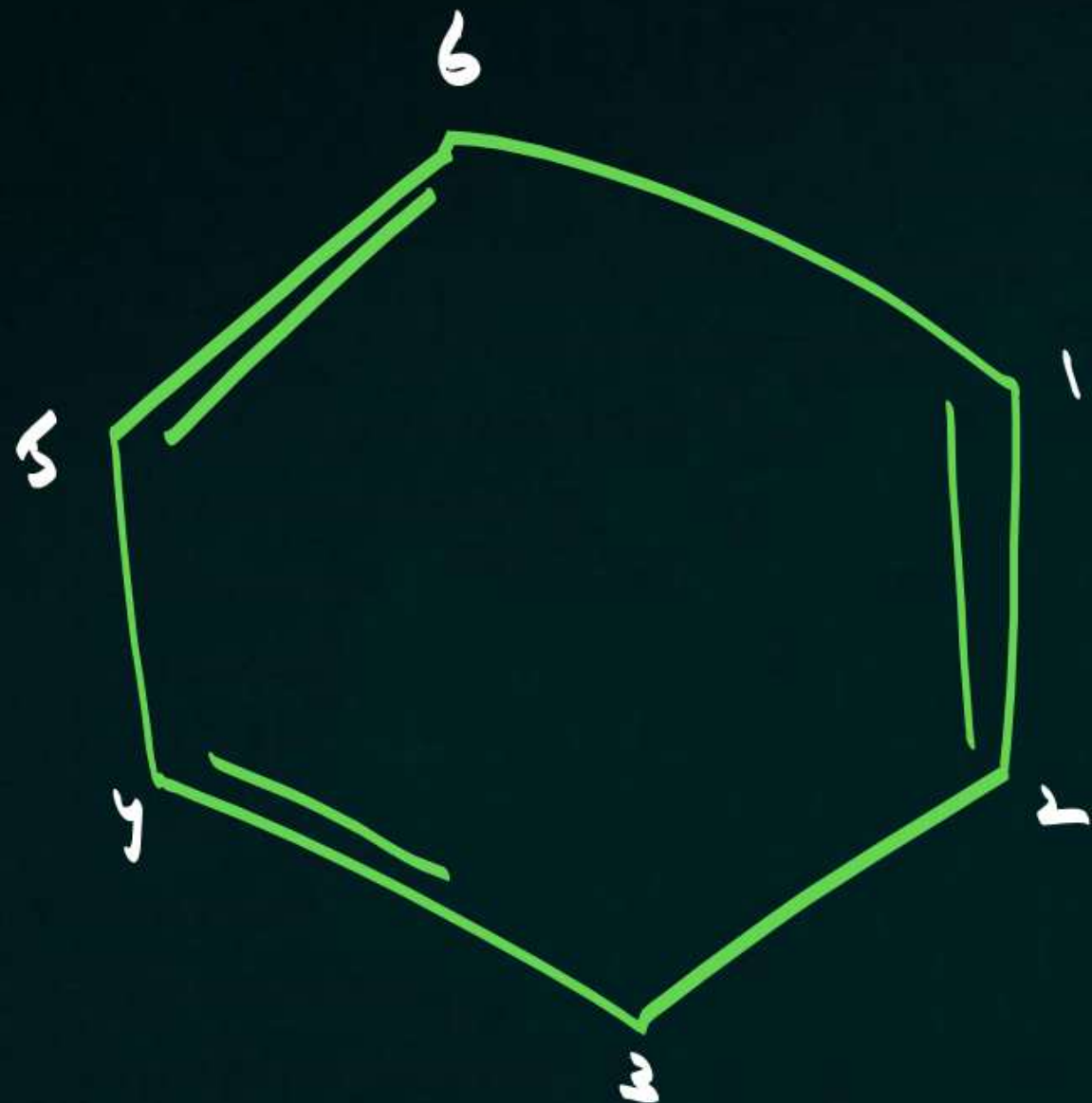
Cyclohex-1-ene.



cyclopent-1-ene



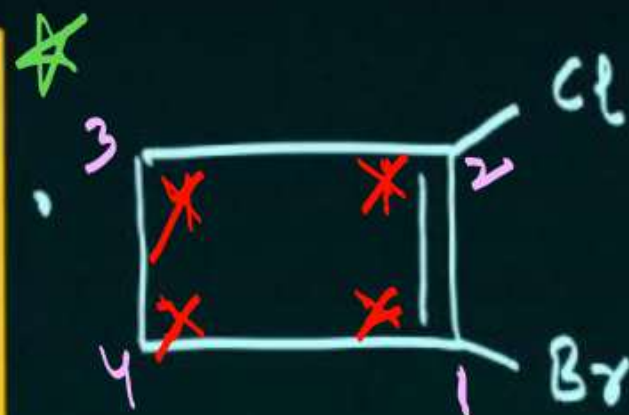
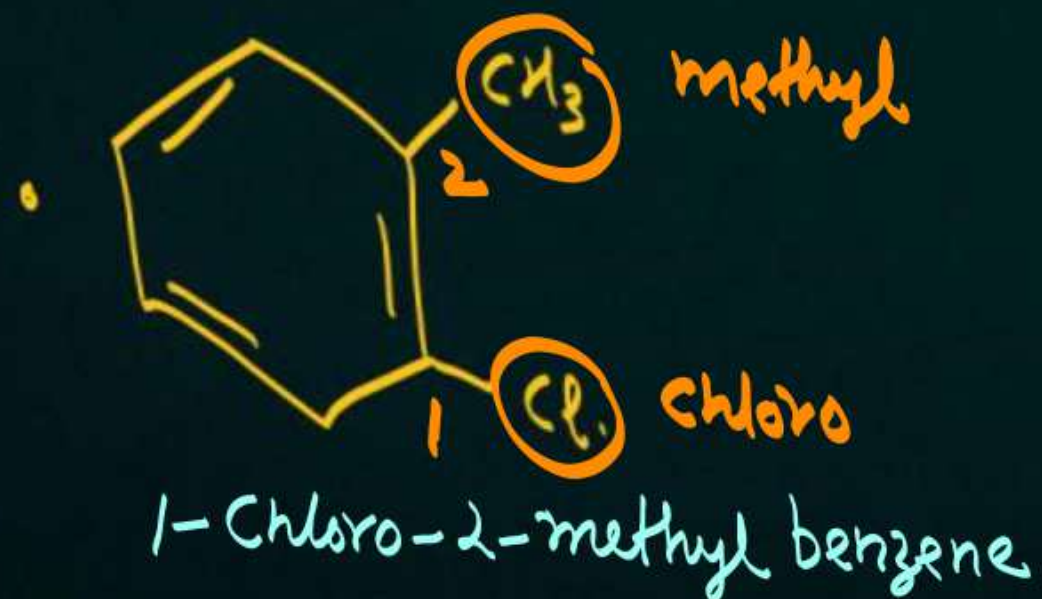
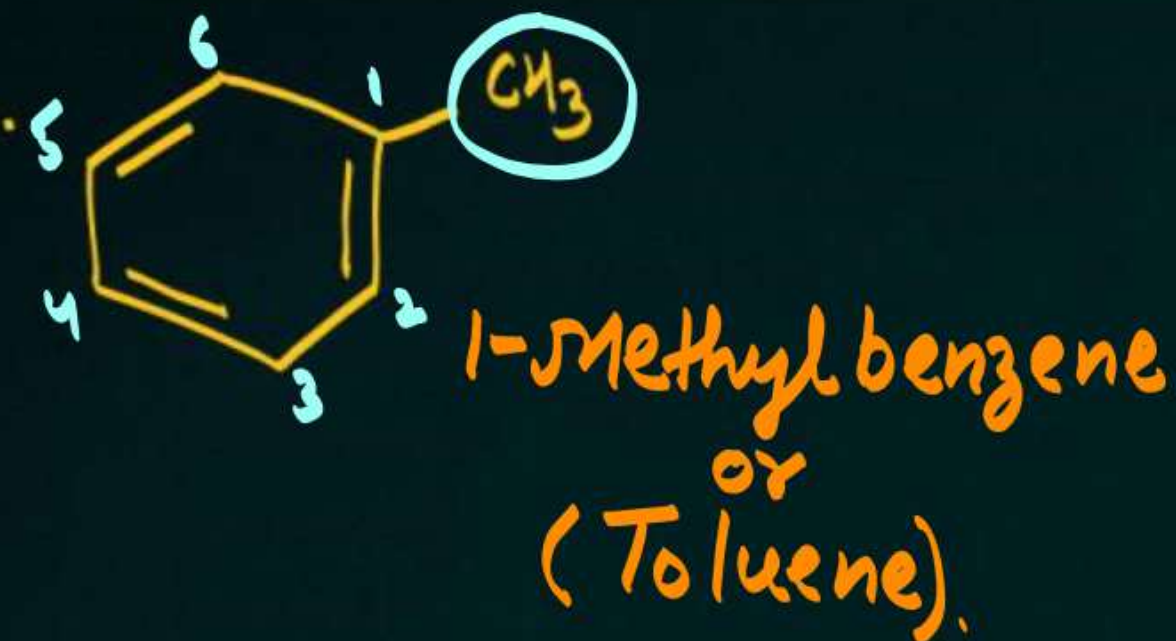
Cyclohex-1,3,5-triene ✗



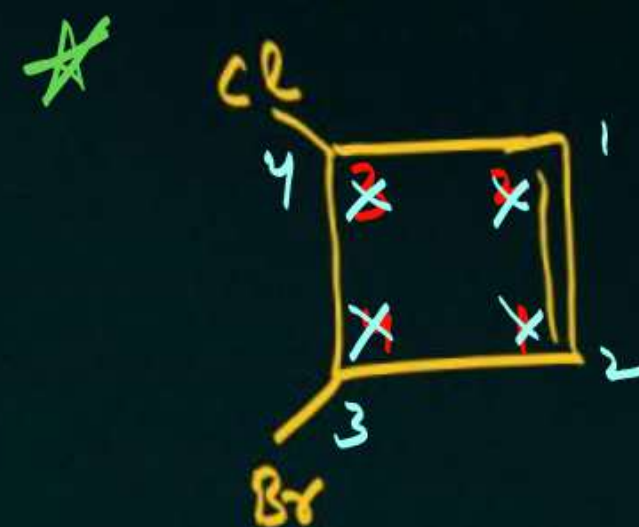
BENZENE
(C_6H_6)

✓

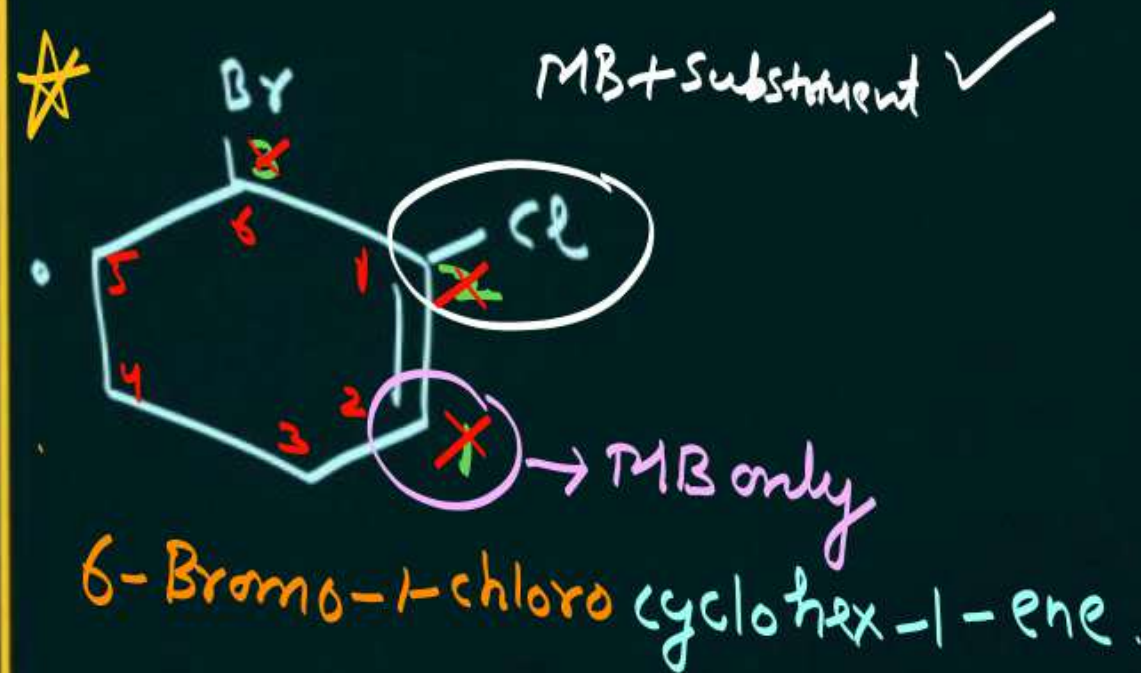
IUPAC Nomenclature



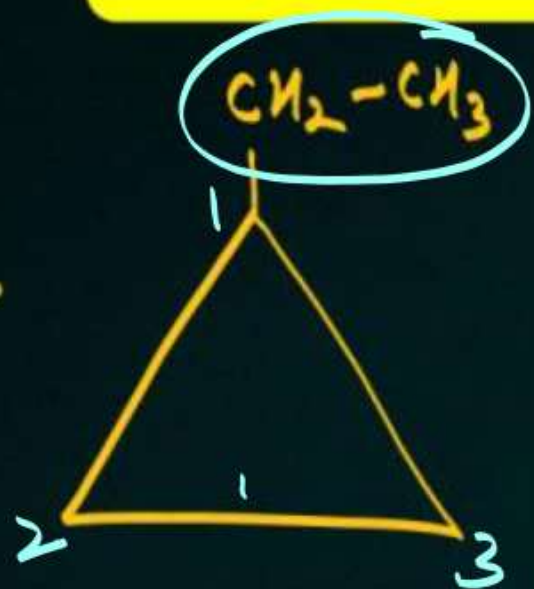
1-Bromo-2-chloro
cyclobut-1-ene



3-Bromo-4-chloro
cyclobut-1-ene.

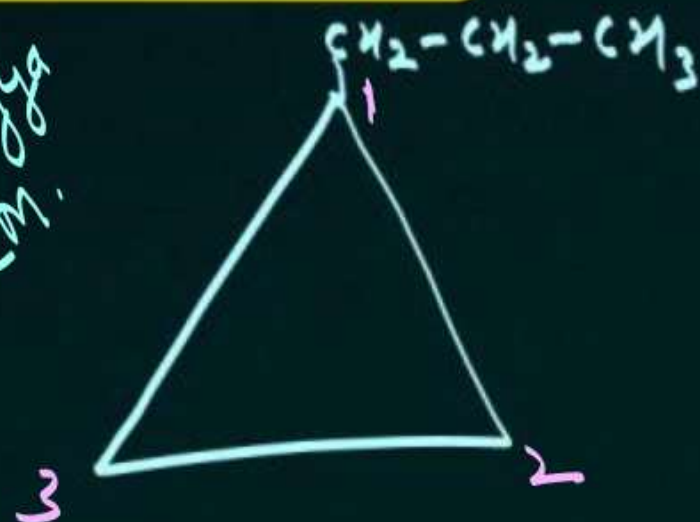


IUPAC Nomenclature



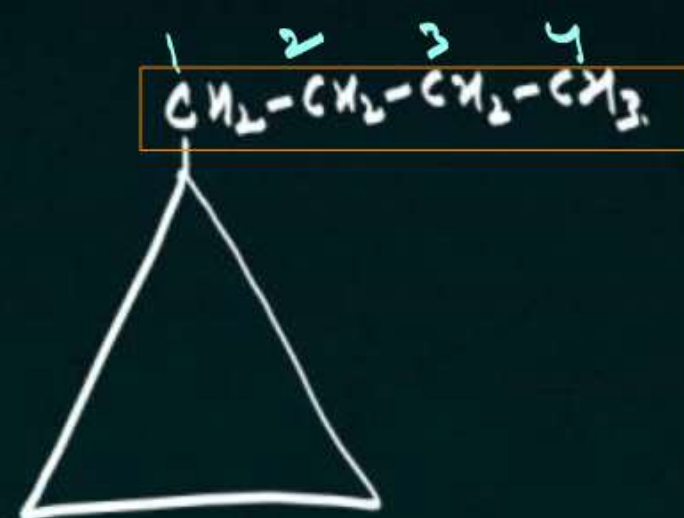
1-Ethylcyclopropane

Fars gya
Cm.

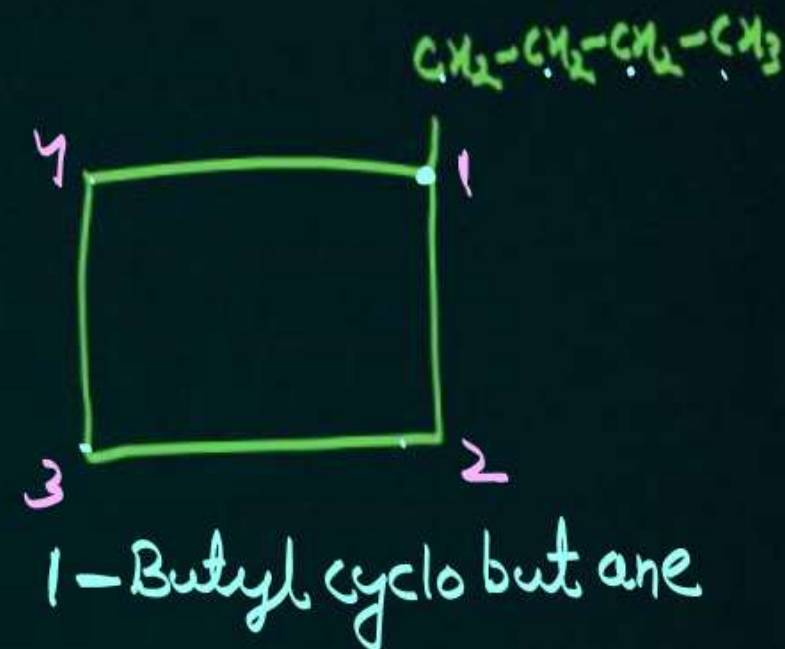


1-Propylcyclopropane

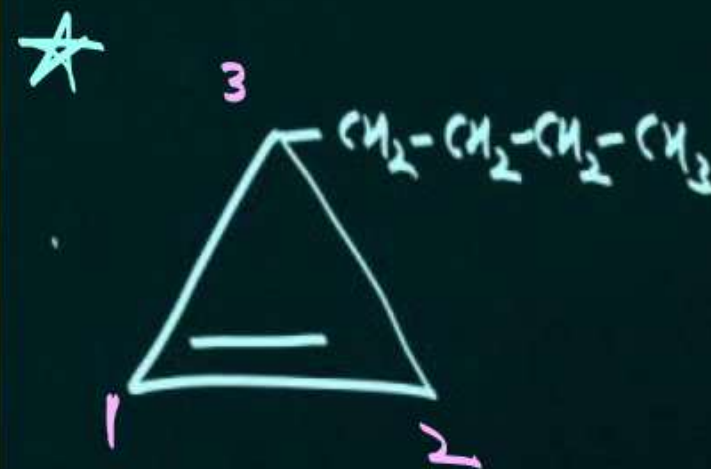
If there are equal no of carbon in linear chain & cyclo alkane then cyclic part is PCC.



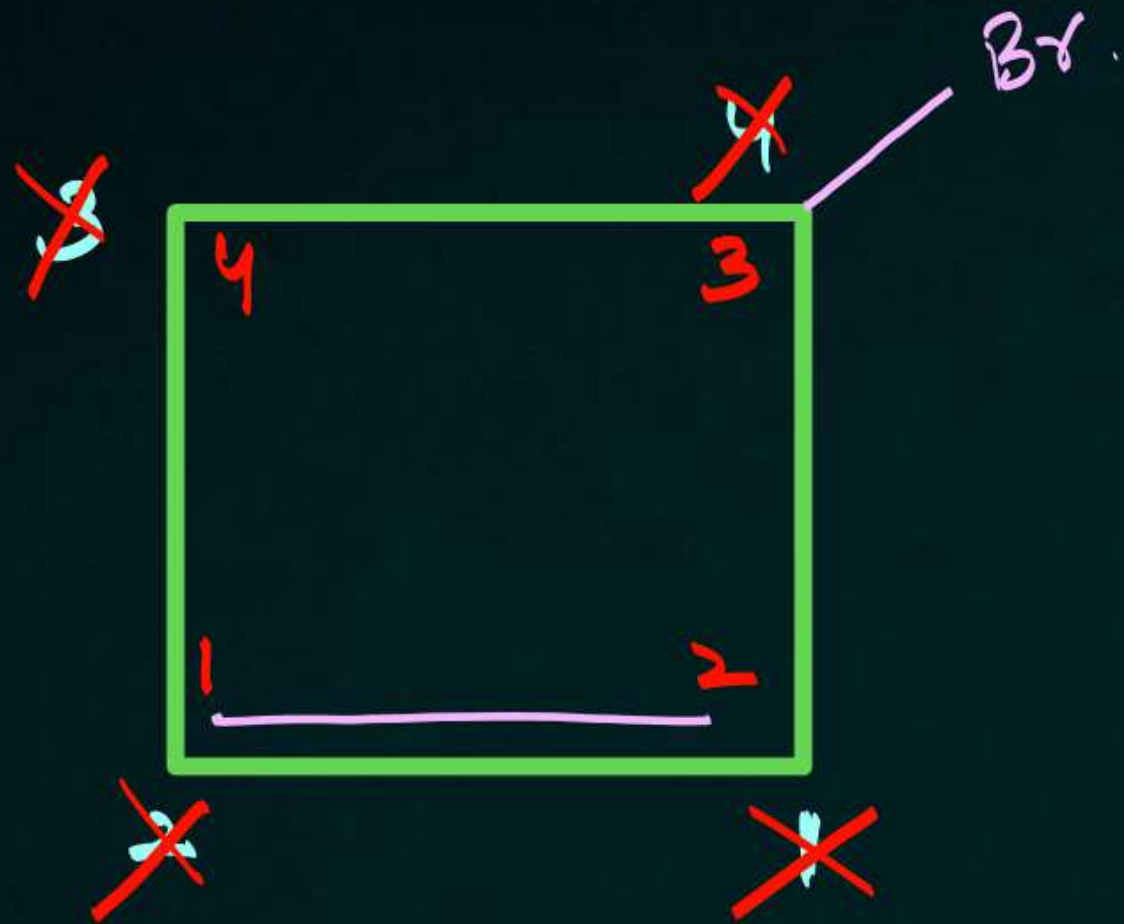
1-Cyclopropylbutane



1-Butylcyclobutane



3-Butylcycloprop-1-ene.



a) 4-Bromo cyclopropene.

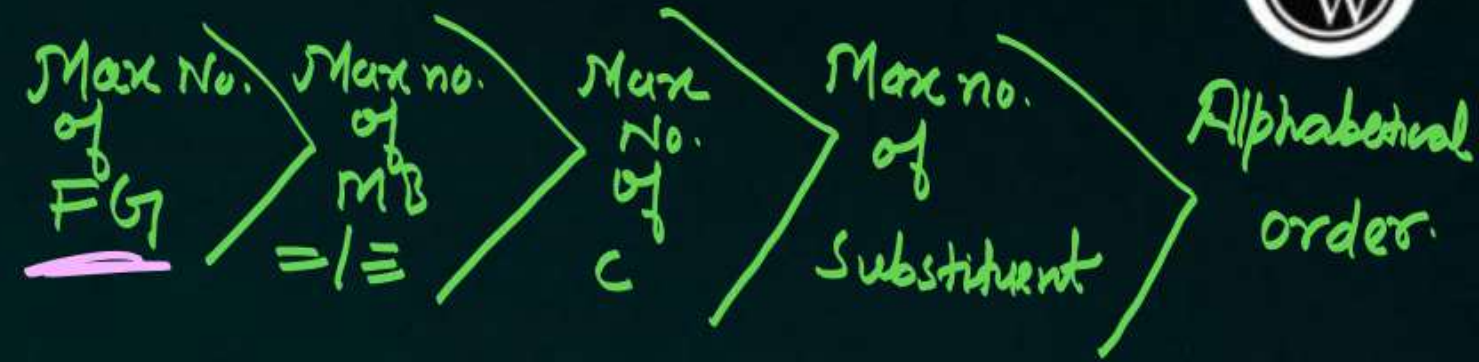
b) 3-Bromocyclobut-1-ene

c) 2-Bromocyclobut-1-ene

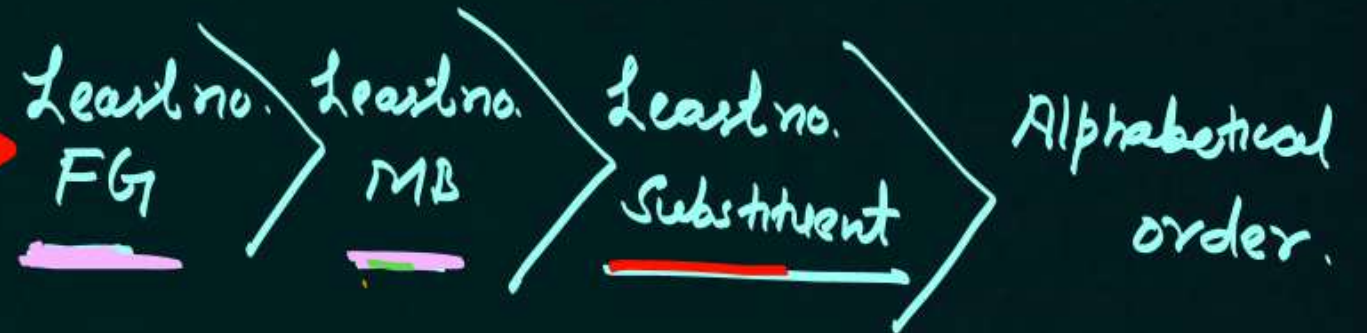
d) 1-Bromocyclobut-2-ene



1) Deciding the PCC



2) NUMBERING OF PCC



3) NAMING

→ SUBSTITUENT + CYCLO + WR + ^{ane} or ^{ene} or ^{yne} + FG.

IUPAC Nomenclature

Subs + cyclo + WR + are or yne + Fd



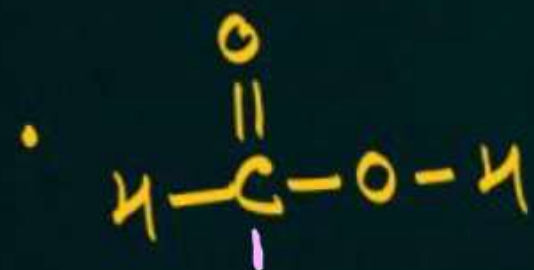
4) NOMENCLATURE OF FUNCTIONAL GROUPS.

$C_n H_{2n} O_2$

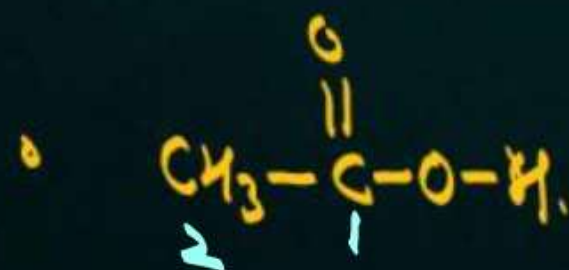
a) Carboxylic acid ($-COOH$), or $-C(=O)OH$

2°-Suffix used = -oic acid

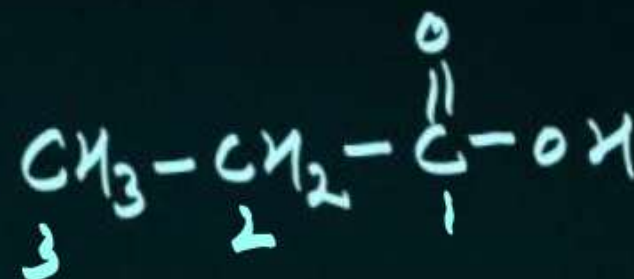
and
ene + Oic acid
yne -



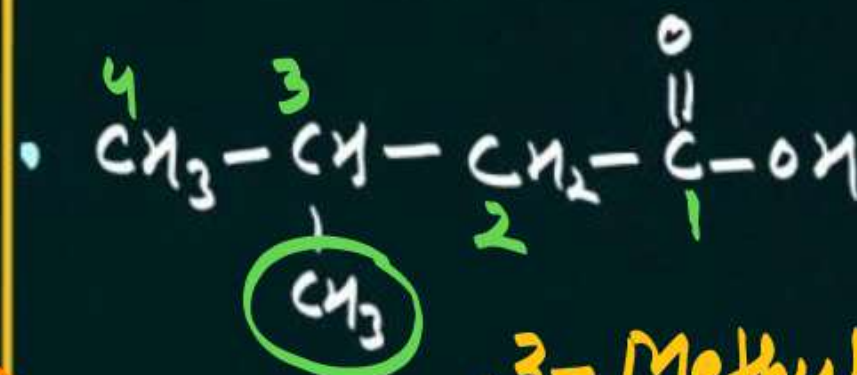
Methanoic acid



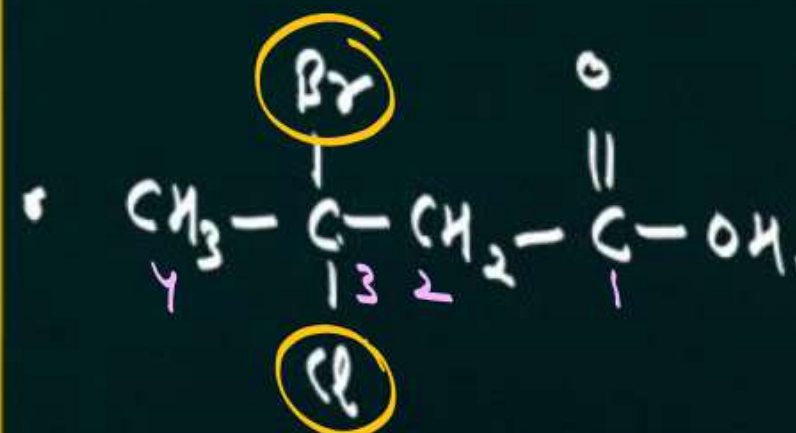
Ethanoic acid



Propanoic acid

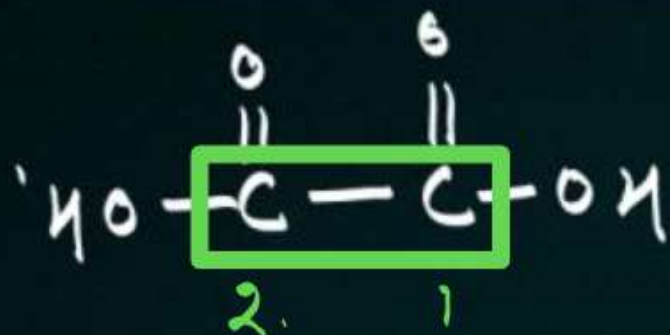


3-Methyl butanoic acid

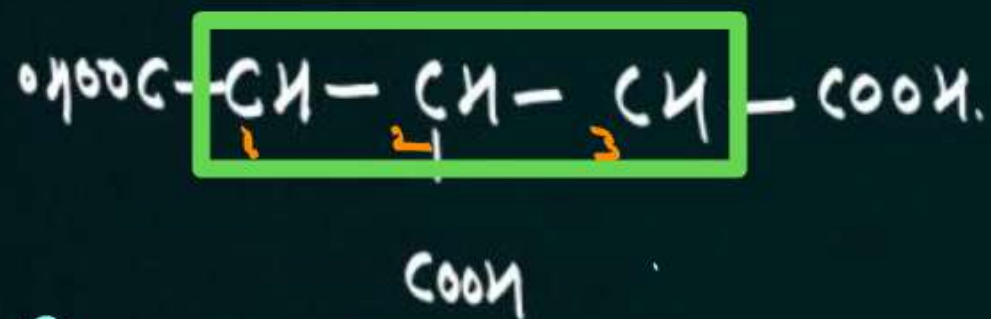


3-Bromo-3-chloro butanoic acid.

IUPAC Nomenclature



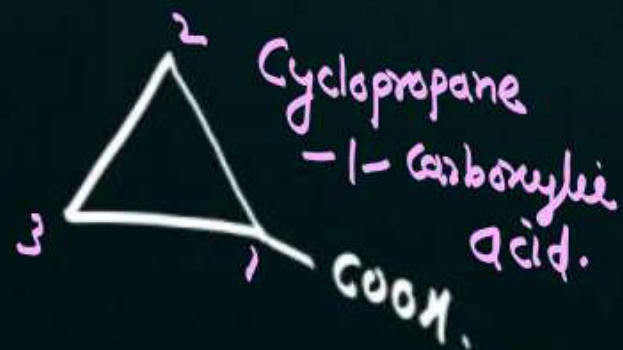
Ethane-1,2-dioic acid



COOH

Propane-1,2,3-Tricarboxylic acid

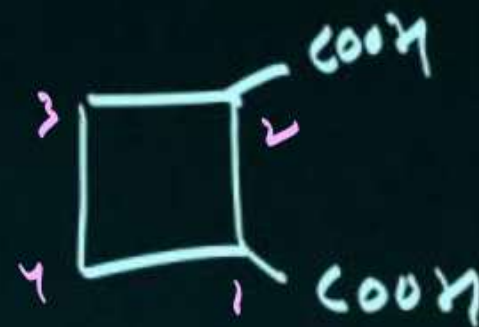
same suffix is used
↑ for cyclic compound.



If in a molecule having more than -COOH group, then if we can't include any one of -COOH group in PC then we omit all the -COOH group and suffix -carboxylic acid is used



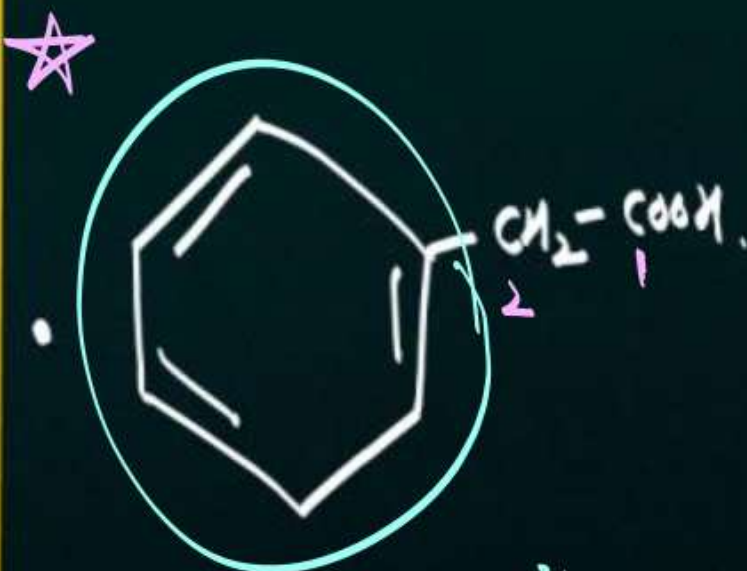
Cyclopentane-1-carboxylic acid



Cyclobutane-1,2-dicarboxylic acid



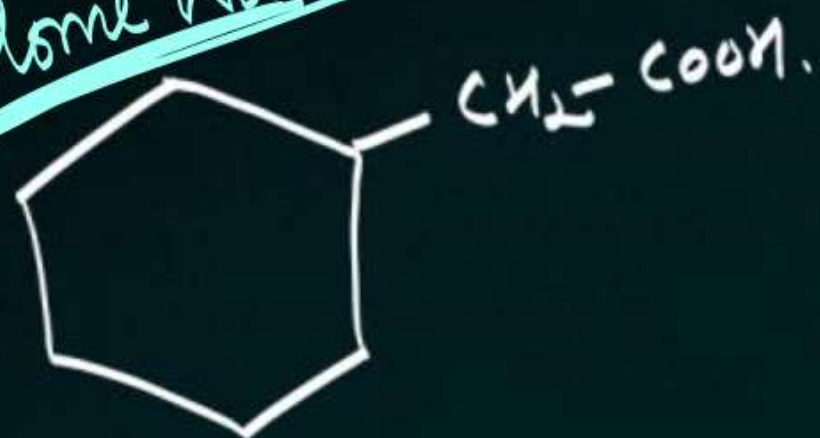
Benzoic acid
or
Benzene carboxylic acid



2-Phenylethan-1-oic acid

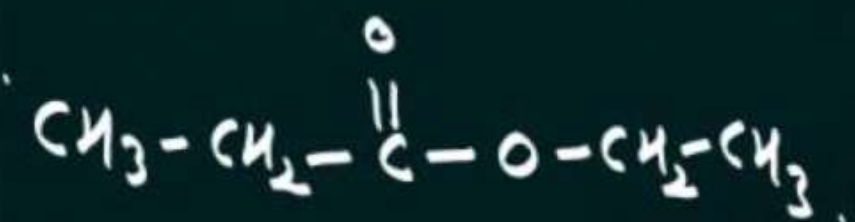
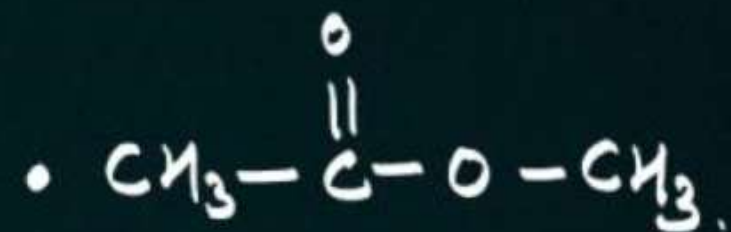
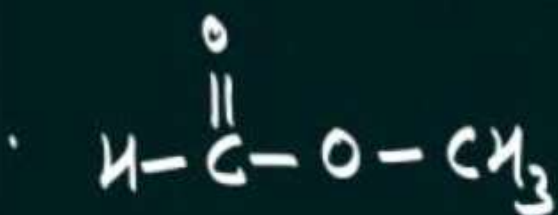
IUPAC Nomenclature

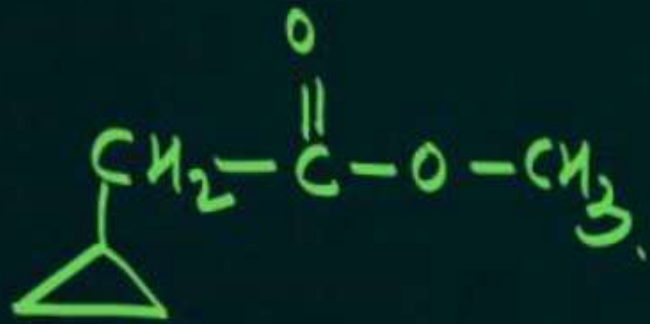
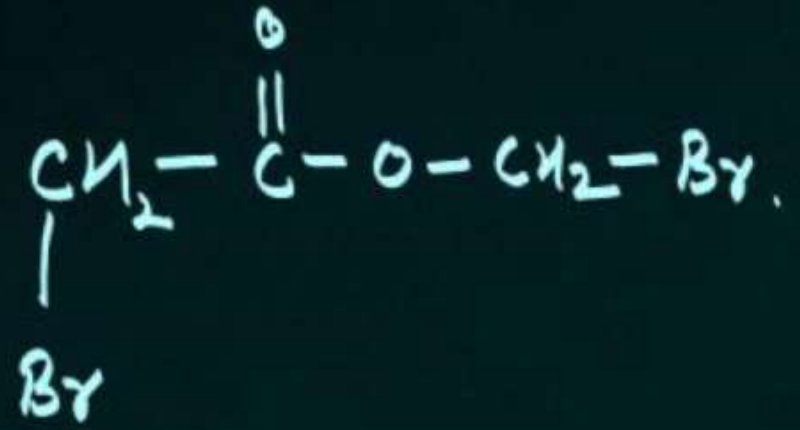
Home Work

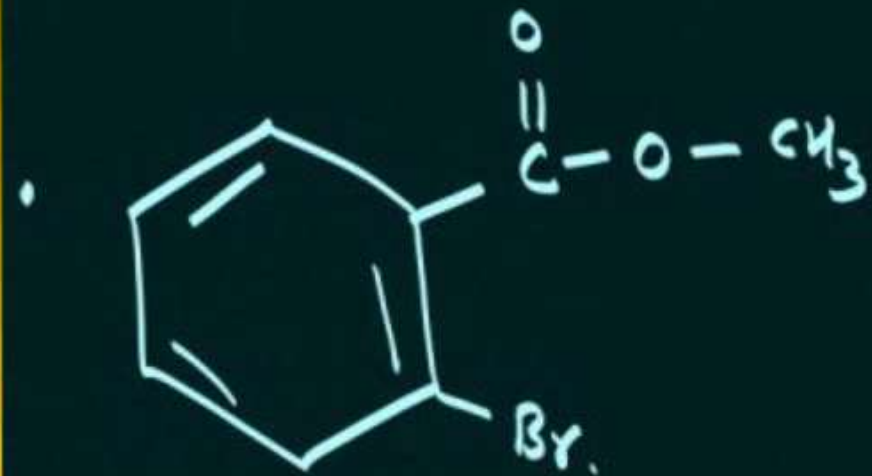
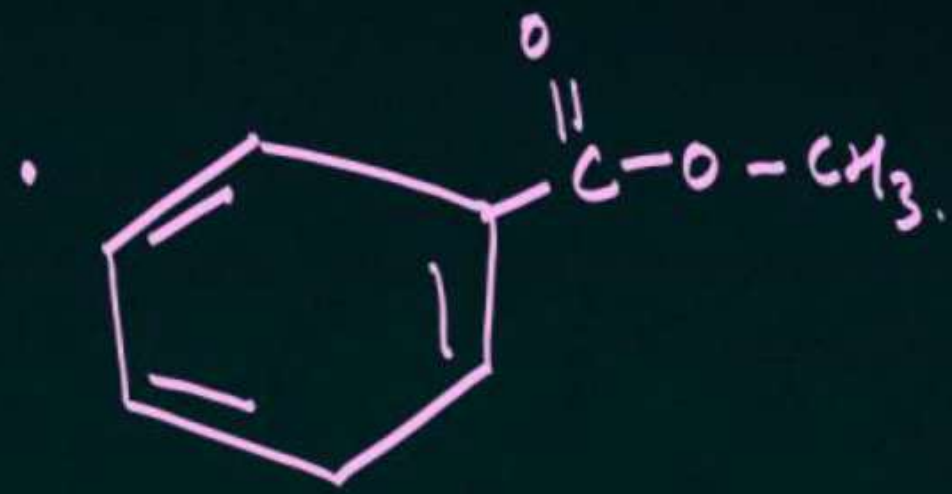
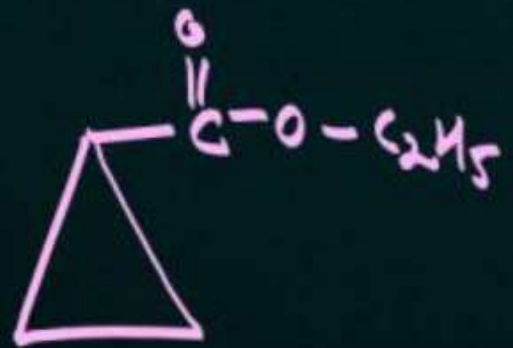
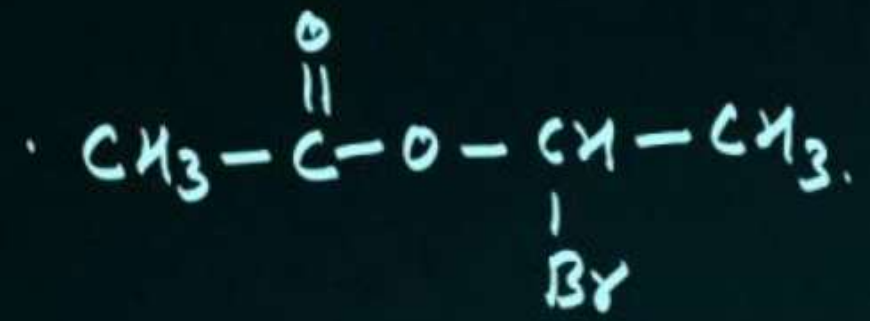


b) Ester $\rightarrow (-\overset{\text{O}}{\parallel}\text{C}-\text{O}-\text{R})$.

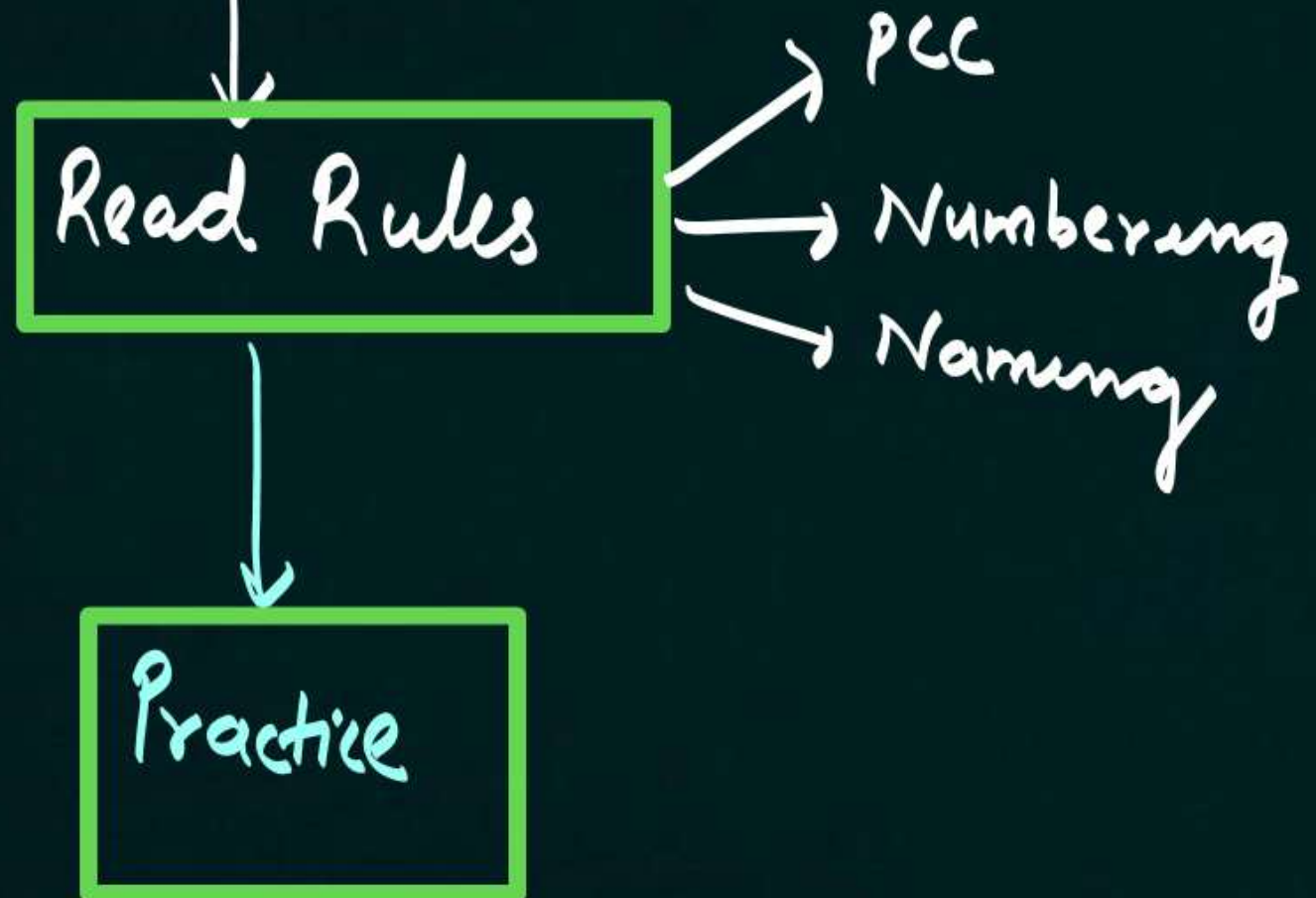
2°-suffix used = -oate







Nomenclature



Sari examples ko Revise Karna.

Thank ~~#XUV~~
You 